RISE AI

Resume, Interview & Skill Enhancement AI

IBM SkillsBuild AICTE Internship Project

Developed by:
Vibhuthi Abhinav

Technology Stack

IBM watsonx Orchestrate

IBM RAG Knowledge Base

Python Flask

HTML

CSS

JavaScript

Acknowledgement

I would like to express my sincere gratitude to IBM SkillsBuild and AICTE for providing me with the opportunity to participate in this internship program and gain practical exposure to Artificial Intelligence and modern software development technologies.

I am grateful to the IBM SkillsBuild learning platform for providing valuable resources, guidance, and hands-on experience with IBM watsonx Orchestrate, Retrieval-Augmented Generation (RAG), and other AI technologies that made the successful completion of this project possible.

I also extend my heartfelt thanks to my faculty members and mentors for their continuous encouragement, valuable suggestions, and support throughout the project development process. Their guidance helped me understand the practical aspects of AI application development and project implementation.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of this project. This internship has significantly enhanced my technical knowledge, problem-solving abilities, and understanding of AI-powered application development.

Abstract

Career preparation has become increasingly challenging due to the growing demand for professional resumes, technical skills, and effective interview performance. To address these challenges, RISE AI (Resume, Interview & Skill Enhancement AI) has been developed as an AI-powered career guidance platform that provides personalized assistance for resume analysis, interview preparation, resume-to-job role matching, skill recommendations, and career guidance.

The application is developed using Python Flask, HTML, CSS, and JavaScript, while its core intelligence is powered by IBM watsonx Orchestrate integrated with a Retrieval-Augmented Generation (RAG) knowledge base. By retrieving relevant information from curated career guidance documents before generating responses, the system provides accurate, context-aware, and reliable recommendations tailored to user requirements.

The project demonstrates the practical implementation of conversational AI and knowledge retrieval in solving real-world career development challenges. The application has been successfully deployed on Render and managed using GitHub, making it easily accessible through a web browser. Additionally, IBM BOB was utilized during the development process to assist in building and refining the frontend and backend components.

Overall, RISE AI serves as a comprehensive and user-friendly career assistance platform that simplifies career preparation while showcasing the capabilities of IBM AI technologies in developing intelligent, scalable, and practical web applications.

Table of Contents

S.NO	Content	Page NO.
1.	Introduction	5
2.	Problem Statement	6
3.	Proposed Solution	7-8
4.	Objectives	9
5.	Features	10-11
6.	System Architecture	12
7.	Workflow	13
8.	Technology Stack	14
9.	Implementation	15
10.	Advantages	16
11.	Results	17
12.	Future Enhancements	18
13.	GitHub Repository and Live Deployment	19
14.	Conclusion	20
15.	References	21

1. Introduction

Artificial Intelligence (AI) is transforming the way people learn, work, and prepare for their careers. In today's competitive job market, students and job seekers need more than technical knowledge—they also require well-structured resumes, interview preparation, and guidance on developing industry-relevant skills. However, many existing career guidance platforms provide only generic advice and lack personalization based on individual user needs.

RISE AI (Resume, Interview & Skill Enhancement AI) is an AI-powered career assistant developed to simplify career preparation through intelligent and personalized guidance. The application integrates **IBM watsonx Orchestrate** with **Retrieval-Augmented Generation (RAG)** to provide accurate, context-aware responses by retrieving information from a curated knowledge base before generating answers. This approach improves the quality and reliability of AI-generated recommendations compared to traditional chatbot systems.

The application is built using **Python Flask** for the backend and **HTML, CSS, and JavaScript** for the frontend, creating a responsive and user-friendly web interface. RISE AI enables users to analyze resumes, compare them with target job roles, prepare for interviews, receive skill recommendations, and ask career-related questions through a single AI-powered platform.

By combining conversational AI with knowledge retrieval, RISE AI demonstrates how modern AI technologies can solve real-world career development challenges. The project provides students and professionals with an accessible tool to improve their employability while showcasing the practical implementation of IBM watsonx Orchestrate and Retrieval-Augmented Generation in a web application.

2. Problem Statement

The recruitment process has become increasingly competitive, requiring candidates to possess not only technical knowledge but also professional resumes, strong interview skills, and relevant industry competencies. Many students and job seekers struggle to prepare resumes that effectively highlight their qualifications, identify the skills required for their target roles, or practice interviews in a structured manner. As a result, they often face difficulties in securing internships and employment opportunities.

Existing career guidance platforms generally focus on a single aspect, such as resume building or interview preparation, requiring users to switch between multiple tools to complete their career preparation. Additionally, many AI chatbots generate generic responses without using verified knowledge sources, which can lead to inconsistent or less relevant recommendations.

There is a need for an intelligent platform that integrates resume analysis, interview preparation, skill enhancement, and career guidance into a single application. Such a system should provide personalized, context-aware responses while ensuring the information is reliable and relevant to the user's queries.

RISE AI addresses these challenges by integrating **IBM watsonx Orchestrate** with a **Retrieval-Augmented Generation (RAG)** knowledge base. The system retrieves relevant information from uploaded career guidance documents before generating responses, enabling users to receive accurate resume feedback, interview guidance, skill recommendations, and career support through a single AI-powered interface. This approach simplifies career preparation while improving the quality and reliability of AI-assisted guidance.

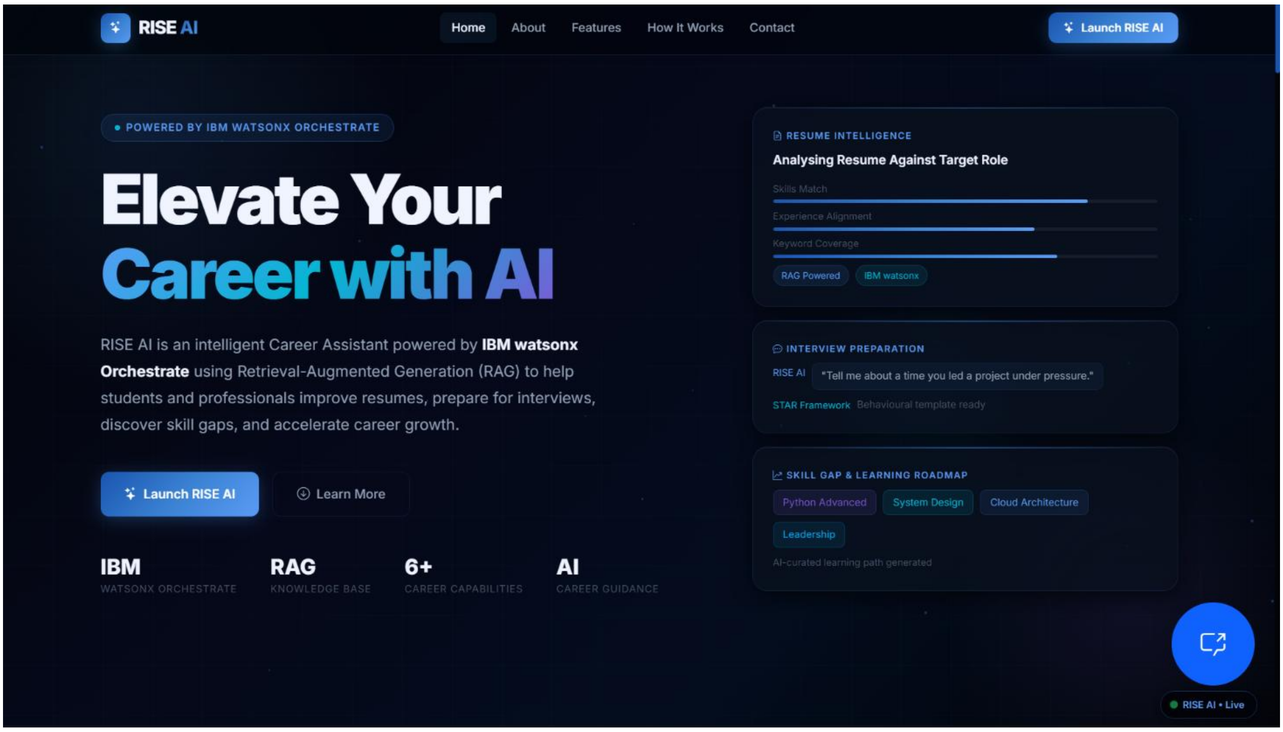
3. Proposed Solution

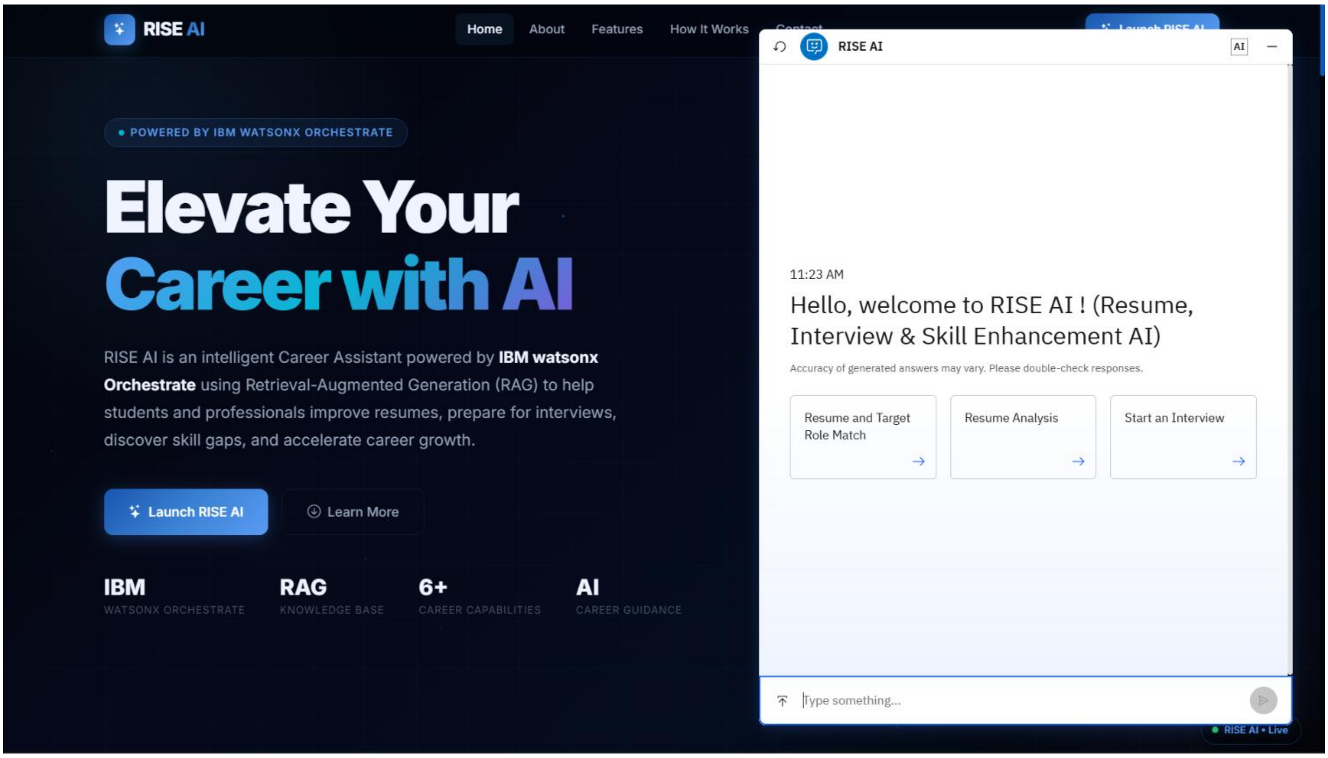
To overcome the limitations of existing career guidance platforms, **RISE AI (Resume, Interview & Skill Enhancement AI)** has been developed as an intelligent web-based application that provides multiple career assistance services through a single platform. The system combines **IBM watsonx Orchestrate** with **Retrieval-Augmented Generation (RAG)** to deliver accurate, personalized, and context-aware responses for career-related queries.

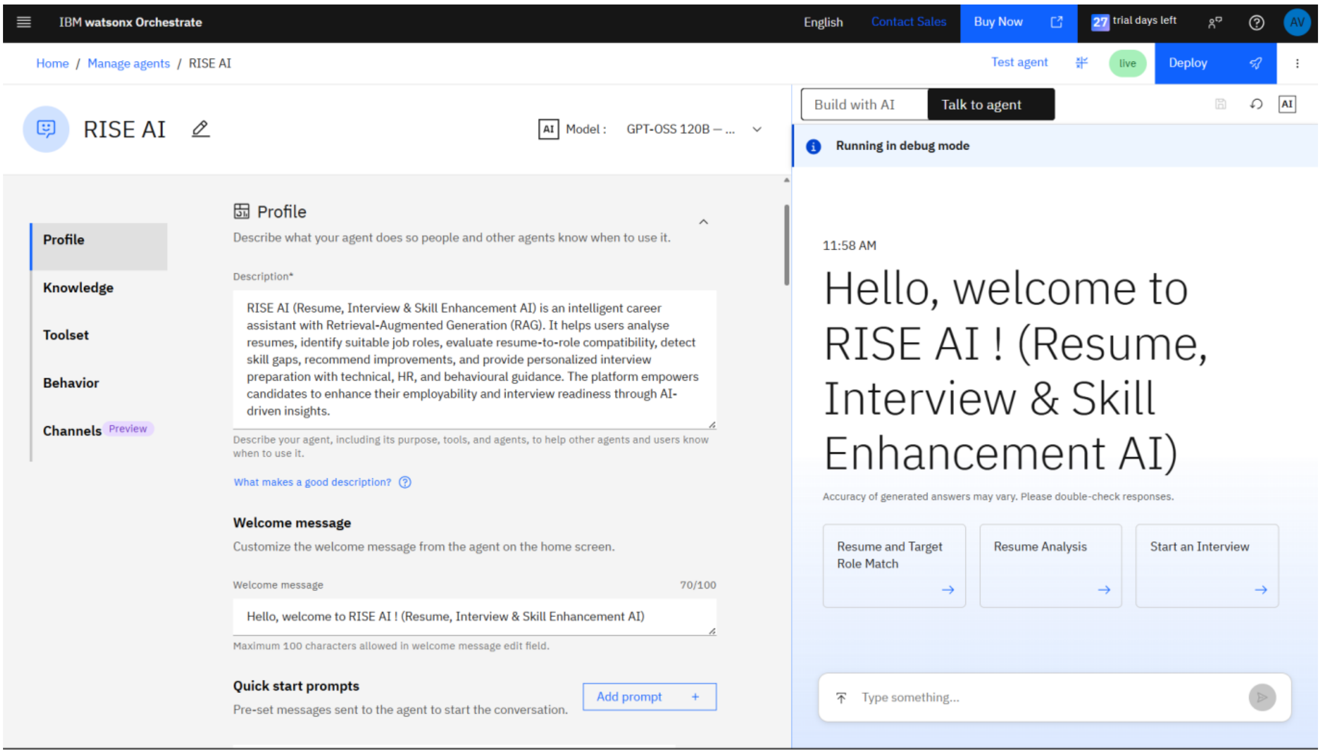
When a user submits a question or uploads a resume, the request is processed by the Flask backend and forwarded to IBM watsonx Orchestrate. The orchestrator retrieves relevant information from the RAG knowledge base, which contains curated career guidance resources such as resume writing techniques, interview preparation materials, and skill development content. Based on the retrieved information, the AI generates a meaningful response and presents it through a simple, user-friendly interface.

The application offers features including resume analysis, resume-to-job role matching, interview question generation, skill recommendations, and AI-powered career guidance. By integrating these functionalities into a single platform, RISE AI eliminates the need to use multiple websites or tools for career preparation.

Overall, the proposed solution demonstrates how AI-powered automation and knowledge retrieval can provide practical career support, helping students and job seekers improve their employability through personalized guidance and accurate information.




RISE AI
Home
About
Features
How It Works
POWERED BY IBM WATSONX ORCHESTRATE
Elevate Your
Career with AI
RISE AI is an intelligent Career Assistant powered by IBM watsonx Orchestrate using Retrieval-Augmented Generation (RAG) to help students and professionals improve resumes, prepare for interviews, discover skill gaps, and accelerate career growth.
Launch RISE AI
Learn More
IBM
WATSONX ORCHESTRATE
RAG
KNOWLEDGE BASE
6+
CAREER CAPABILITIES
AI
CAREER GUIDANCE
RISE AI
11:23 AM
Hello, welcome to RISE AI ! (Resume, Interview & Skill Enhancement AI)
Accuracy of generated answers may vary. Please double-check responses.
Resume and Target Role Match
Resume Analysis
Start an Interview
Type something...
RISE AI • Live


IBM watsonx Orchestrate
English
Contact Sales
Buy Now
27 trial days left
Home / Manage agents / RISE AI
Test agent
live
Deploy
RISE AI
Model : GPT-OSS 120B – ...
Build with AI
Talk to agent
Running in debug mode
Profile
Knowledge
Toolset
Behavior
Channels Preview
Profile
Describe what your agent does so people and other agents know when to use it.
Description*
RISE AI (Resume, Interview & Skill Enhancement AI) is an intelligent career assistant with Retrieval-Augmented Generation (RAG). It helps users analyse resumes, identify suitable job roles, evaluate resume-to-role compatibility, detect skill gaps, recommend improvements, and provide personalized interview preparation with technical, HR, and behavioural guidance. The platform empowers candidates to enhance their employability and interview readiness through AI-driven insights.
Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.
What makes a good description?
Welcome message
Customize the welcome message from the agent on the home screen.
Welcome message
70/100
Hello, welcome to RISE AI ! (Resume, Interview & Skill Enhancement AI)
Maximum 100 characters allowed in welcome message edit field.
Quick start prompts
Pre-set messages sent to the agent to start the conversation.
Add prompt +
11:58 AM
Hello, welcome to RISE AI ! (Resume, Interview & Skill Enhancement AI)
Accuracy of generated answers may vary. Please double-check responses.
Resume and Target Role Match
Resume Analysis
Start an Interview
Type something...

4. Objectives

The primary objective of **RISE AI** is to develop an intelligent AI-powered career assistant that helps students and job seekers improve their career readiness through personalized guidance. The application aims to simplify the career preparation process by providing multiple services within a single platform, reducing the need for separate tools and resources.

The specific objectives of the project are:

- To analyze user resumes and provide meaningful suggestions for improvement.
- To compare a resume with a target job role and identify areas that need enhancement.
- To generate interview questions and preparation guidance based on the user's desired role.
- To recommend relevant technical and soft skills that improve employability.
- To provide accurate and context-aware responses using **IBM watsonx Orchestrate** integrated with a **Retrieval-Augmented Generation (RAG)** knowledge base.
- To develop a responsive and user-friendly web application using **Python Flask**, **HTML**, **CSS**, and **JavaScript**.
- To demonstrate the practical application of AI and knowledge retrieval technologies in solving real-world career development challenges.

By achieving these objectives, RISE AI serves as a comprehensive career guidance platform that supports users in resume building, interview preparation, skill enhancement, and informed career planning.

5. Features

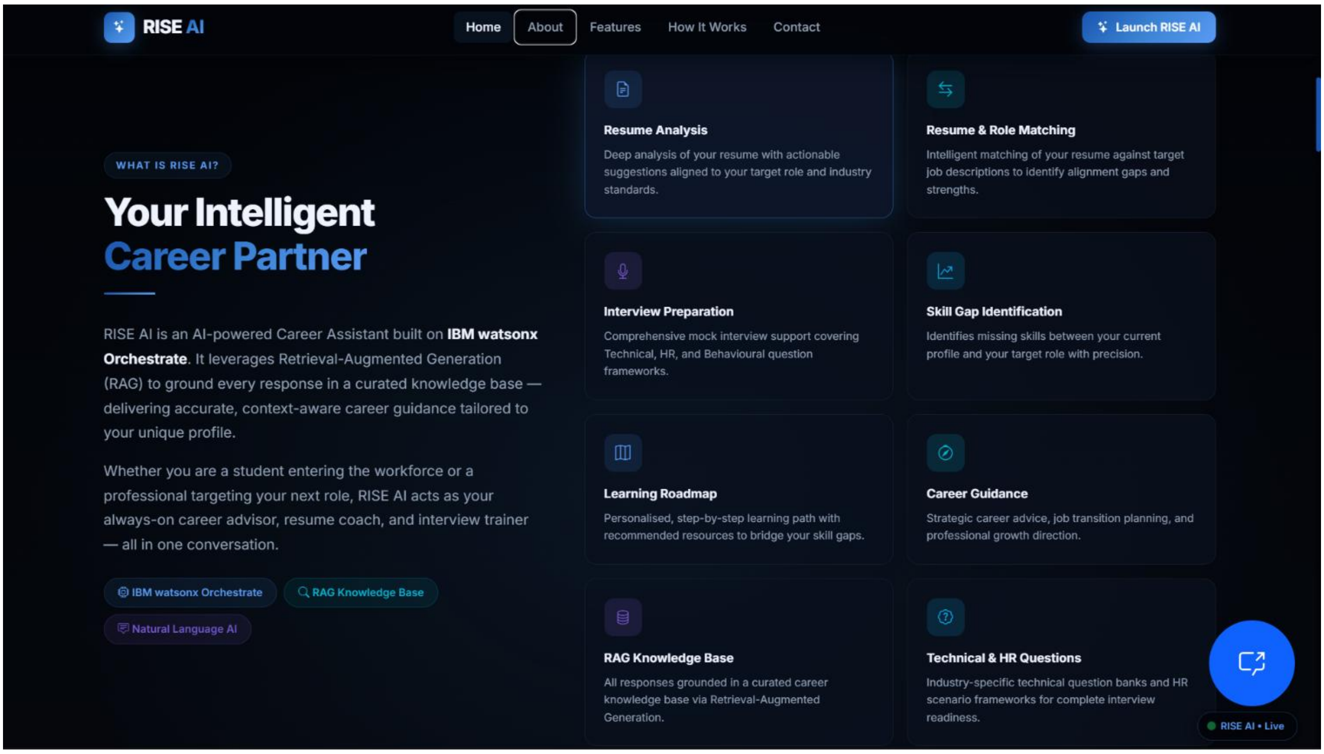
RISE AI provides a comprehensive set of AI-powered features designed to support students and job seekers throughout their career preparation journey. The application combines multiple career assistance services into a single platform, making it convenient and efficient for users.

The key features of RISE AI include:

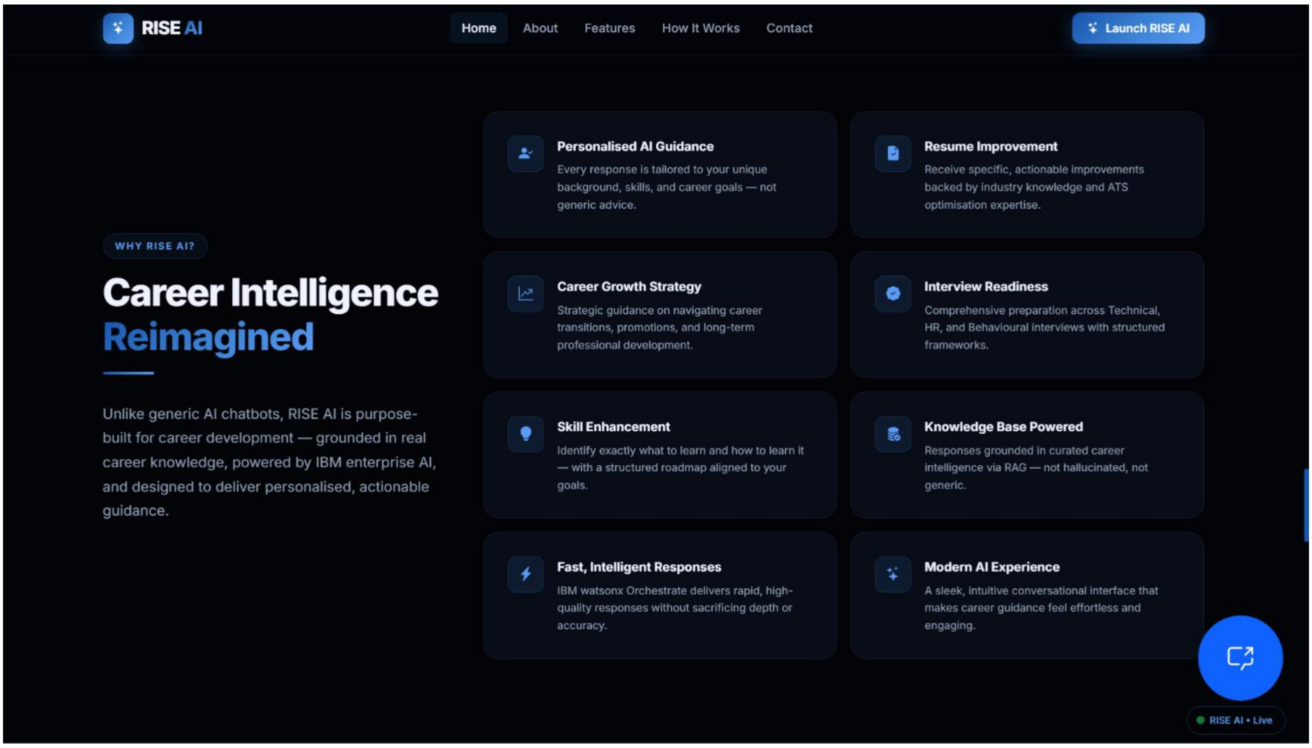
- **Resume Analysis:** Evaluates uploaded resumes and provides suggestions to improve structure, content, and overall quality.
- **Resume–Job Role Matching:** Compares a user's resume with a target job role to identify skill gaps and areas for improvement.
- **Interview Preparation:** Generates role-specific interview questions and helps users prepare for technical and HR interviews.
- **Skill Recommendations:** Suggests relevant technical and soft skills based on career goals and industry requirements.
- **AI Career Assistant:** Answers career-related questions and provides personalized guidance through a conversational interface.
- **Retrieval-Augmented Generation (RAG):** Retrieves information from a curated knowledge base before generating responses, ensuring accurate and context-aware recommendations.
- **Responsive Web Interface:** Offers a clean, intuitive, and user-friendly interface accessible through modern web browsers.

These features work together to provide an end-to-end career guidance solution, helping users improve their resumes, strengthen interview preparation, enhance professional skills, and make informed career decisions.

What is RISE AI:



WHY RISE AI:

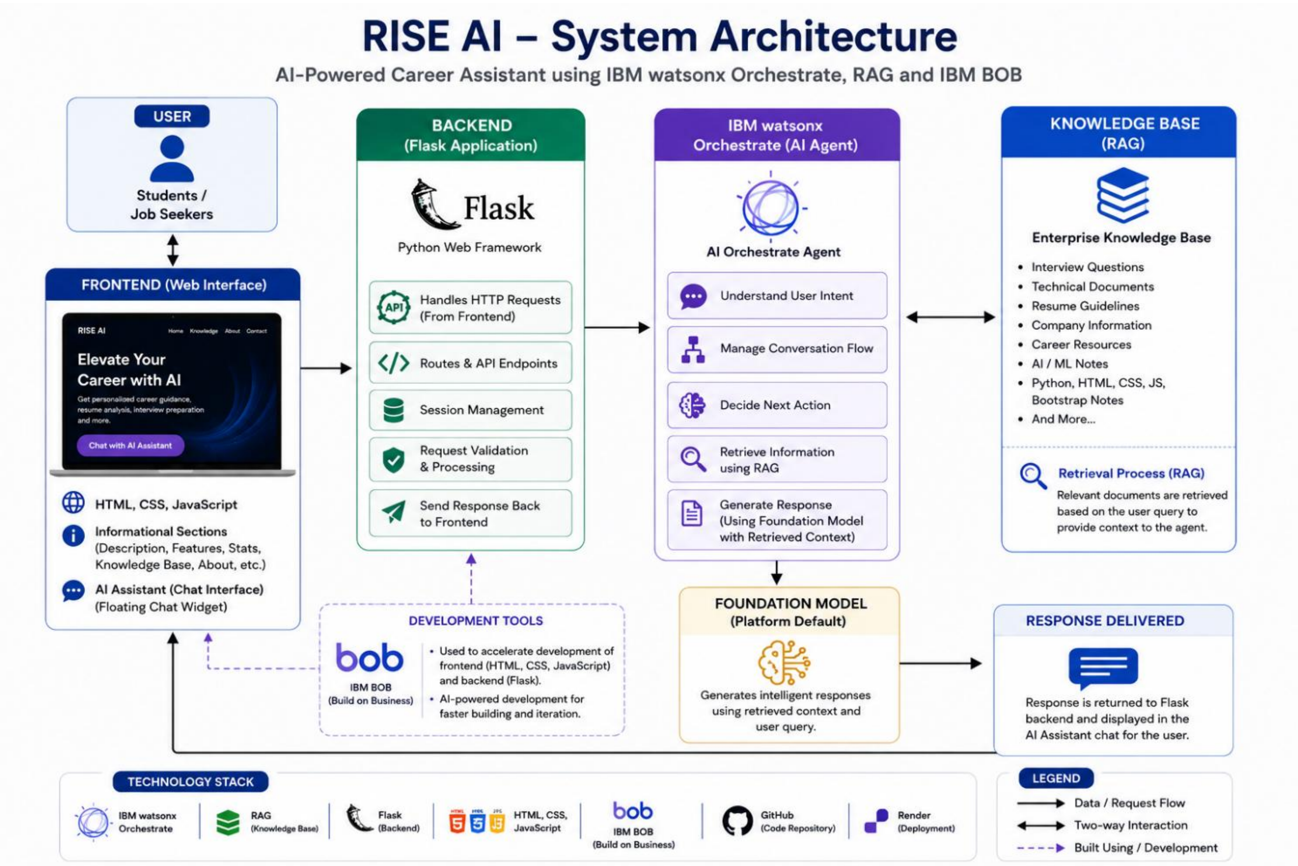


6. System Architecture

The architecture of **RISE AI** is designed to provide seamless communication between the user, the web application, and the AI services. It follows a modular architecture where each component performs a specific function, ensuring efficient processing of user requests and accurate response generation.

The process begins when a user submits a query or uploads a resume through the web interface developed using **HTML, CSS, and JavaScript**. The request is then sent to the **Python Flask** backend, which manages the application logic and communicates with **IBM watsonx Orchestrate**. The orchestrator processes the request and retrieves relevant information from the **RAG Knowledge Base**, which contains curated career guidance resources. Using the retrieved information, the AI generates a contextual response and sends it back to the Flask application. Finally, the processed response is displayed to the user through the web interface.

This modular architecture ensures scalability, maintainability, and efficient integration of AI services while providing users with fast and reliable career guidance.

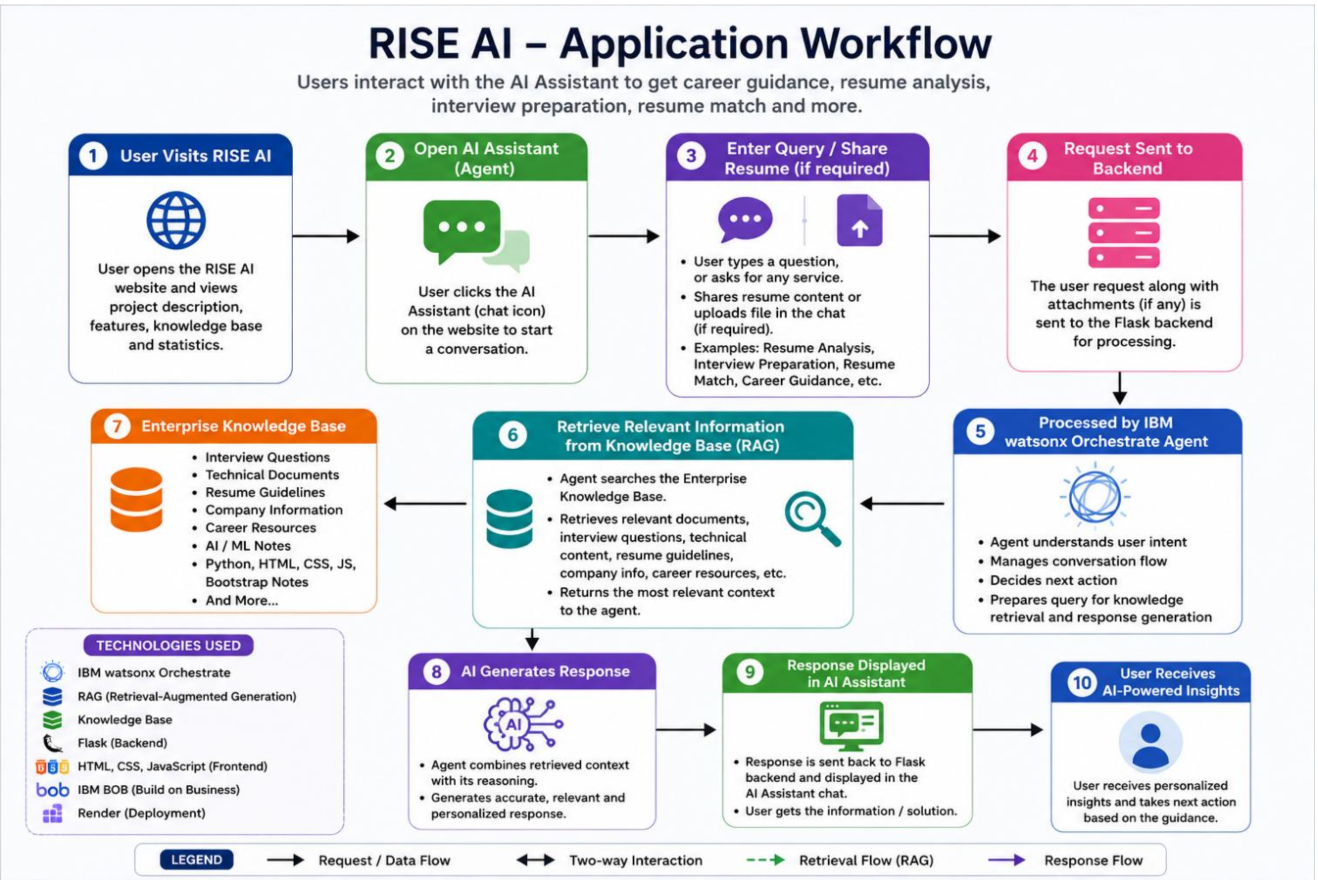


7. Workflow

The workflow of **RISE AI** illustrates how a user request is processed from submission to response generation. The application follows a structured sequence that integrates the web interface, backend processing, IBM watsonx Orchestrate, and the RAG knowledge base to deliver accurate and personalized career guidance.

The process begins when a user opens the application and submits a career-related query or uploads a resume. The request is received by the **Python Flask** backend, which validates the input and forwards it to **IBM watsonx Orchestrate**. The orchestrator then searches the **Retrieval-Augmented Generation (RAG) Knowledge Base** to retrieve relevant career guidance information. Based on the retrieved content, the AI generates a context-aware response and sends it back to the Flask application. Finally, the response is displayed on the web interface, allowing the user to review resume suggestions, interview guidance, skill recommendations, or answers to career-related questions.

This workflow ensures efficient request processing, reliable knowledge retrieval, and an interactive user experience while maintaining the accuracy and relevance of AI-generated responses.



8. Technology Stack

RISE AI is developed using a combination of modern web technologies and IBM AI services to provide a reliable, scalable, and user-friendly career guidance platform. Each technology plays a specific role in the overall system architecture and functionality.

The **frontend** is built using **HTML, CSS, and JavaScript**, providing a responsive and intuitive interface for user interaction. The **backend** is developed using **Python Flask**, which manages application logic, processes user requests, and communicates with IBM watsonx Orchestrate. The AI capabilities are powered by **IBM watsonx Orchestrate**, which coordinates the processing of user queries and generates intelligent responses. To improve the accuracy and relevance of responses, the application uses a **Retrieval-Augmented Generation (RAG) Knowledge Base** containing curated career guidance documents. The project source code is managed using **GitHub**, and the web application is deployed on **Render**, enabling users to access the system online through a web browser.

Component	Technology Used	Purpose
Frontend	HTML, CSS, JavaScript	User Interface
Backend	Python Flask	Application Logic & Request Handling
AI Platform	IBM watsonx Orchestrate	AI Response Generation
Knowledge Base	RAG	Context-Aware Information Retrieval
Version Control	GitHub	Source Code Management
Deployment	Render	Web Application Hosting
AI Development Assistant	IBM BOB	Assisted in developing and refining the frontend and backend components

9. Implementation

The implementation of **RISE AI** follows a modular approach, separating the application into frontend, backend, AI integration, and deployment components. This structure improves maintainability, scalability, and ease of future enhancements.

The **frontend** is developed using **HTML, CSS, and JavaScript**, providing a clean and responsive interface where users can submit career-related queries or upload resumes. The **Python Flask** backend handles routing, processes user requests, manages communication between the frontend and IBM watsonx Orchestrate, and returns AI-generated responses to the user interface.

The core intelligence of the application is powered by **IBM watsonx Orchestrate**, which processes user requests and retrieves relevant information from the **Retrieval-Augmented Generation (RAG) Knowledge Base**. This enables the system to generate accurate and context-aware responses for resume analysis, interview preparation, skill recommendations, and career guidance.

After development and testing, the application was uploaded to **GitHub** for version control and deployed on **Render**, making it accessible through a web browser. The deployment ensures that users can interact with the application from any device with an internet connection.

The modular implementation allows the system to be easily updated with additional features, making RISE AI a scalable and practical AI-powered career guidance platform.

During the development process, **IBM BOB** was used as an AI-assisted development tool to accelerate the implementation of the frontend and backend components. It provided guidance for code generation, debugging, interface design, and integration, enabling faster development while maintaining the overall architecture based on IBM watsonx Orchestrate, RAG, and Python Flask.

10. Advantages

RISE AI offers several advantages by combining Artificial Intelligence with Retrieval-Augmented Generation (RAG) to provide a personalized and efficient career guidance experience. Unlike traditional career preparation tools that focus on a single functionality, RISE AI integrates multiple services into one platform, making career preparation more convenient and effective for users.

The application provides intelligent resume analysis, resume-to-job role matching, interview preparation, skill recommendations, and AI-powered career guidance through a simple and interactive web interface. The integration of **IBM watsonx Orchestrate** with a **RAG Knowledge Base** ensures that responses are context-aware, accurate, and based on curated knowledge rather than relying solely on the language model. This improves the reliability and quality of the recommendations provided to users.

Additionally, the web-based architecture allows the application to be accessed from any device with an internet connection, eliminating the need for complex installations. The modular design also makes the system easy to maintain, update, and extend with future features.

Key Advantages

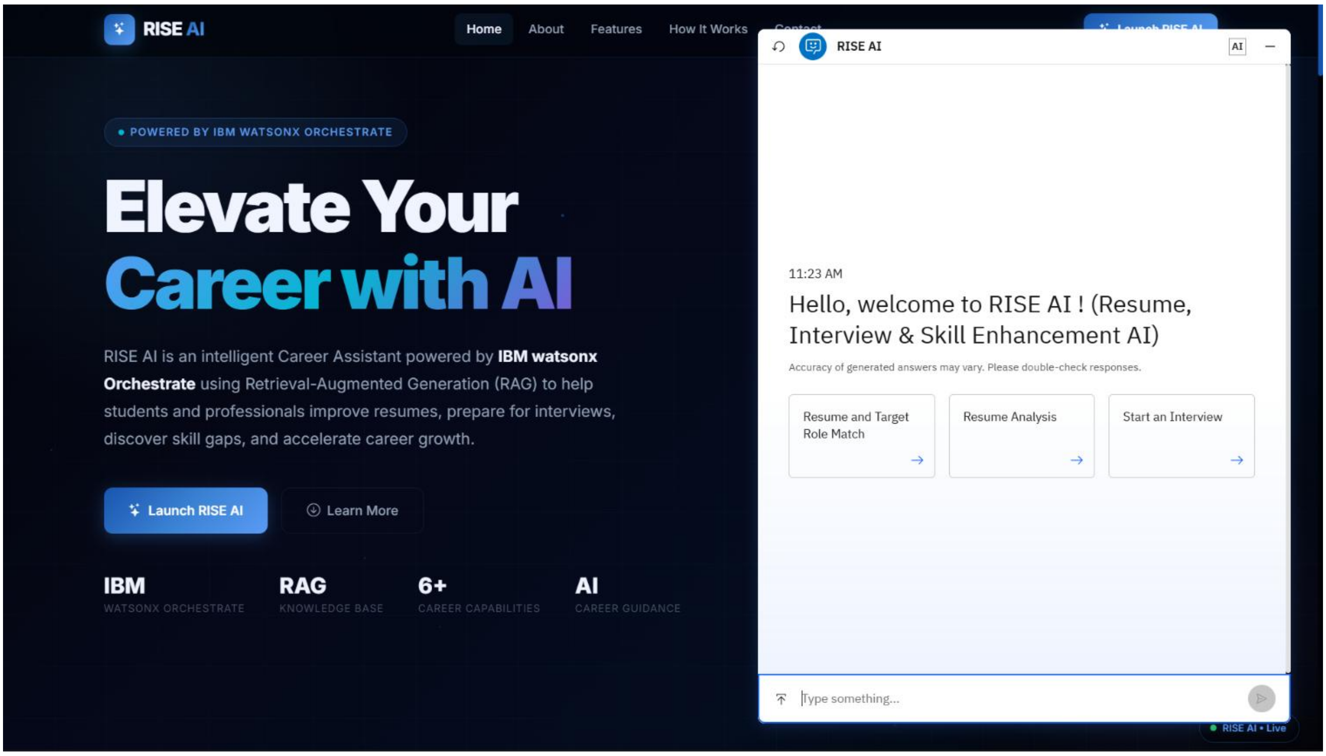
- Provides personalized career guidance through AI.
- Analyzes resumes and suggests meaningful improvements.
- Compares resumes with target job roles to identify skill gaps.
- Generates interview preparation guidance and role-specific questions.
- Delivers accurate responses using Retrieval-Augmented Generation (RAG).
- Combines multiple career development services into a single platform.
- User-friendly, responsive, and easily accessible through a web browser.
- Scalable architecture that supports future enhancements and integrations.

11. Results

The developed **RISE AI** application was successfully implemented and tested, demonstrating its ability to provide intelligent and personalized career guidance through an interactive web interface. The system effectively processes user queries, communicates with **IBM watsonx Orchestrate**, retrieves relevant information from the **RAG Knowledge Base**, and generates accurate, context-aware responses.

During testing, the application successfully performed key functionalities such as resume analysis, resume-to-job role matching, interview question generation, skill recommendations, and AI-powered career assistance. The integration of Retrieval-Augmented Generation improved the relevance of responses by using information from curated knowledge documents instead of relying only on the language model. The application also provided a smooth and responsive user experience through its web interface and remained accessible after deployment on Render.

The successful implementation demonstrates that RISE AI can serve as a practical career guidance solution for students and job seekers by combining AI-driven automation with knowledge retrieval to deliver reliable and personalized recommendations.



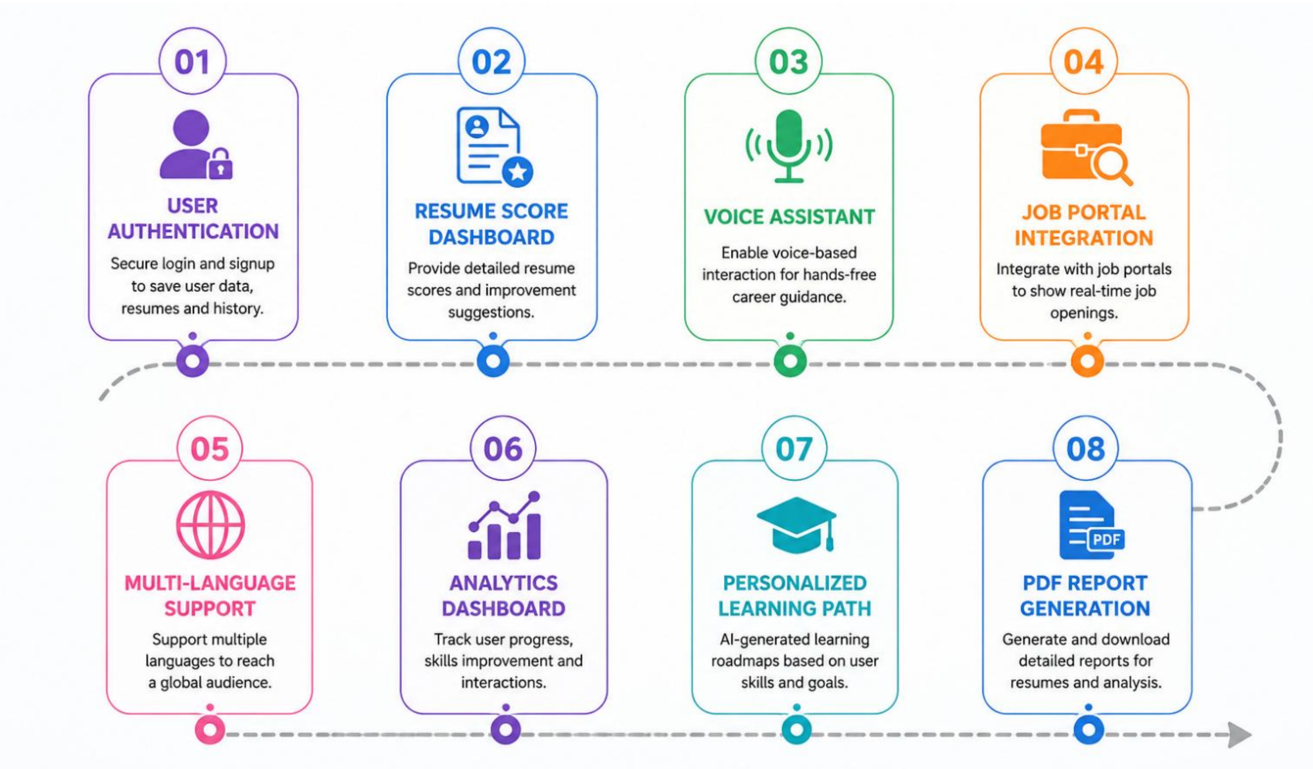
12. Future Enhancements

Although RISE AI successfully provides AI-powered career guidance, the application can be further enhanced by adding new features and improving its overall functionality. Future versions of the project can focus on increasing personalization, expanding AI capabilities, and integrating additional career development services.

Some of the planned enhancements include implementing **user authentication** to allow users to securely store their resumes and access previous interactions. A **resume scoring dashboard** can be added to provide detailed performance metrics and track improvements over time. The application can also support **multiple languages**, making it accessible to a wider range of users.

Additional features such as **voice-based AI interaction**, **PDF report generation**, and **integration with job portals** can further improve the user experience. Personalized learning roadmaps, progress tracking, and analytics dashboards can also help users monitor their career development more effectively. As AI technologies continue to evolve, newer IBM AI services and advanced language models can be integrated to provide even more accurate and intelligent career guidance.

These enhancements will make RISE AI a more comprehensive, scalable, and user-centric platform capable of supporting a broader range of career development needs.



13. GitHub Repository and Live Deployment

To ensure accessibility, maintainability, and version control, the RISE AI project is hosted on GitHub and deployed as a live web application using Render. GitHub serves as the central repository for managing the project source code, tracking updates, and maintaining different versions of the application. It also enables collaboration and provides easy access to the project's implementation files and documentation.

The application is deployed on the Render cloud platform, allowing users to access RISE AI directly through a web browser without requiring local installation or configuration. The deployment demonstrates the successful integration of the frontend, backend, and IBM watsonx Orchestrate services into a fully functional online application. Hosting the project online improves accessibility, simplifies demonstration, and showcases the practical deployment of an AI-powered web application.

GitHub Repository:
https://github.com/Abhinav00010/RISE-AI

Live Application:

https://rise-ai-pm3b.onrender.com

Please note: The application is hosted on the free tier of Render. If the application has been inactive, the first request may take **30–60 seconds** to process while the server starts. During this time, please wait without refreshing the page. Once the server is active, subsequent requests will be processed much faster.

The successful deployment confirms that RISE AI is a production-ready web application capable of delivering AI-powered career guidance through a responsive and user-friendly interface.

14. Conclusion

RISE AI is an AI-powered career guidance platform developed to assist students and job seekers in improving their employability through intelligent and personalized support. By integrating **IBM watsonx Orchestrate** with a **Retrieval-Augmented Generation (RAG)** knowledge base, the application delivers accurate, context-aware responses for resume analysis, interview preparation, skill recommendations, and career-related queries.

The project demonstrates the practical application of Artificial Intelligence in solving real-world career development challenges. The use of **Python Flask** for backend development and **HTML, CSS, and JavaScript** for the frontend provides a responsive and user-friendly web interface, while deployment on **Render** ensures easy online accessibility. The integration of a curated knowledge base improves the reliability of AI-generated responses, making the platform more effective than traditional chatbot-based systems.

Overall, RISE AI successfully achieves its objective of providing a centralized and intelligent career assistance platform. The project showcases the capabilities of IBM AI technologies in building practical, scalable, and user-focused applications. With future enhancements such as user authentication, analytics, multilingual support, and job portal integration, RISE AI has the potential to evolve into a comprehensive career development solution that benefits students, fresh graduates, and professionals alike.

15. References

[1] IBM, "IBM watsonx Orchestrate Documentation." [Online]. Available: https://www.ibm.com/docs/en/watsonx/watson-orchestrate. [Accessed: July , 2026].

[2] IBM, "IBM watsonx Orchestrate Developer Documentation." [Online]. Available: https://developer.watson-orchestrate.ibm.com/. [Accessed: July , 2026].

[3] IBM, "IBM BOB IDE Documentation." [Online]. Available: https://bob.ibm.com/docs/ide. [Accessed: July , 2026].

[4] IBM, "IBM SkillsBuild." [Online]. Available: https://skillsbuild.org/. [Accessed: July, 2026].

[5] Pallets Projects, "Flask Documentation." [Online]. Available: https://flask.palletsprojects.com/. [Accessed: July, 2026].

[6] Mozilla Developer Network (MDN), "HTML Documentation." [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/HTML. [Accessed: July, 2026].

[7] Mozilla Developer Network (MDN), "CSS Documentation." [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/CSS. [Accessed: July, 2026].

[8] Mozilla Developer Network (MDN), "JavaScript Documentation." [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/JavaScript. [Accessed: July, 2026].

[9] GitHub, "GitHub Documentation." [Online]. Available: https://docs.github.com/. [Accessed: July, 2026].

[10] Render, "Render Documentation." [Online]. Available: https://render.com/docs. [Accessed: July, 2026].

[11] IBM, "What is Retrieval-Augmented Generation (RAG)?", [Online]. Available: https://www.ibm.com/think/topics/retrieval-augmented-generation. [Accessed: Jul. 7, 2026].